

BRIEF REPORT

Stereotactic Radiation Therapy or Protons for Uveal Melanoma Patients? An Artificial Intelligence (AI)-Based Clinical Treatment Decision-Making Tool Predicting Doses To Radiation Therapy Constraints



Emmanuelle Fleury, MSc,^{a,b} Jean-Philippe Pignol, MD, PhD,^c Emine Kiliç, MD, PhD,^{d,e} Caroline van Rij, MD,^a Nicole Naus, MD, PhD,^d Serdar Yavuziyigitoglu, MD, PhD,^d Wilhelm den Toom, MSc,^a Andras Zolnay, PhD,^a Kees Spruijt, MSc,^b Marco van Vulpen, MD, PhD,^b Petra Trnková, PhD,^{a,f} and Mischa Hoogeman, PhD^{a,b}

^aDepartment of Radiotherapy, Erasmus Medical Center Cancer Institute, University Medical Center, Rotterdam, The Netherlands;

^bHolland Proton Therapy Center, Delft, The Netherlands; ^cSultan Qaboos Comprehensive Cancer and Research Center, Muscat,

Oman; ^dDepartment of Ophthalmology, Erasmus Medical Center, Rotterdam, The Netherlands; ^eDepartment of Clinical Genetics,

Erasmus Medical Center, Rotterdam, The Netherlands; and ^fDepartment of Radiation Oncology, Medical University of Vienna, Vienna, Austria

Received Nov 28, 2023; Revised May 18, 2025; Accepted for publication Jun 17, 2025

Purpose: For ocular melanoma, selecting between stereotactic radiation therapy (SRT) and protons requires a lengthy plan comparison process. The purpose of this brief report is to describe an artificial intelligence (AI) decision-making tool to predict dosimetric and clinical outcomes based on easy-to-access tumor characteristics.

Methods and Materials: The AI tool was based on a retrospective database of 66 patients with uveal melanoma treated in a single center with robotic SRT. A supervised machine learning model was developed to correlate the risk of toxicity for each radiation modality and clinical features. Clinical toxicity risks were built in various profiles: Profile I for maculopathy, optic-neuropathy, and visual acuity deterioration; Profile II for neovascular glaucoma; Profile III for radiation-induced retinopathy; and Profile IV for dry-eye syndrome.

Corresponding author: Emmanuelle Fleury, MSc; E-mail: e.fleury@erasmusmc.nl

Author Responsible for Statistical Analysis: Emmanuelle Fleury.

Disclosures: This research was co-funded by the research program PROTONS4Vision (Grant NWO 14654), which was financed by the Netherlands Organization for Scientific Research (NWO), Technology Foundation STW, the Top consortium for Knowledge & Innovation (TKI-HTSM), and Varian, a Siemens Healthineers Company, Palo Alto, California, USA. None of the funding organizations had any influence on the design or conduct of the study. E.F. has research collaborations with Elekta AB, Accuray Inc, Varian, and RaySearch Laboratories outside the submitted work. J.-P. P. was senior vice president, chief medical and technology officer at Accuray Inc, until February 2023. M.H. reports grants from the Netherlands Organization for Scientific Research, from Varian (a Siemens Healthineers Company, Palo Alto, California, USA) during the conduct of the study. He is a member of advisory board of Accuray (Sunnyvale, USA); a participant/

presenter at Accuray Thinktank Meeting on Prostate cancer, outside the submitted work. The Department of Radiotherapy (Erasmus Medical Center Cancer Institute, The Netherlands) has research collaborations with Elekta AB, Accuray Inc, Varian, RaySearch Laboratories outside the submitted work. The other authors have nothing to disclose.

Data Sharing Statement: Research data are stored in an institutional repository and will be shared upon request to the corresponding author.

Acknowledgments—This research was co-funded by the research program PROTONS4Vision (Grant NWO 14654), which was financed by the Netherlands Organization for Scientific Research (NWO), Technology Foundation STW, the Top consortium for Knowledge & Innovation (TKI-HTSM), and Varian, a Siemens Healthineers Company, Palo Alto, California, USA.

Supplementary material Table E1 associated with this article can be found in the online version at [doi:10.1016/j.ijrobp.2025.06.3867](https://doi.org/10.1016/j.ijrobp.2025.06.3867).

Results: Machine learning-based toxicity prediction accuracy for selecting the correct treatment modality was 81%, 77%, 91%, and 93% for Profiles I, II, III, and IV, respectively.

Conclusion: The study shows that a machine learning method based on easy-to-access clinical characteristics can predict which toxicity would be greater with SRT or protons. This AI tool could support patients in making informed treatment decisions in an ophthalmology clinic, without the lengthy wait for computed tomographic simulation results and extensive plan comparisons. © 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>)

Introduction

Uveal melanoma (UM) is a rare malignancy frequently treated with radiation therapy to preserve the eye.¹ Techniques include ocular proton therapy,²⁻⁴ external beam stereotactic radiation therapy (SRT),^{5,6} and episcleral brachytherapy plaques.⁷ Local tumor control is excellent for all techniques, so the choice between techniques should be based on the risk of treatment complications.⁸ However, those modalities cannot be compared side-by-side because published series do not include the same patients' characteristics regarding tumor size, location, or pre-existing medical conditions. Those techniques use different imaging for planning,⁹ and the treatment volumes, dose, and fractionation are also different. There is currently no consensus or decision aid helping to choose between SRT or protons, and the only way is to compare side-by-side dose distributions for an individual patient, which is a time-consuming process requiring in-depth expertise.

The aim of this study was to utilize artificial intelligence (AI) to assist in decision-making during the initial ophthalmic oncology consultation, and following ultrasound evaluation, by identifying the most appropriate radiation therapy technique based on the patient's specific tumor characteristics. To this end, an AI-based decision aid tool was developed correlating the clinical characteristics of the tumor with the dosimetric differences between SRT and protons.

Materials and Methods

The study population included all patients with UM treated with robotic SRT at Erasmus Medical Center, The Netherlands, between 2016 and 2021. Details on the study population can be found in Fleury et al.¹⁰ The study was approved by the Institutional Ethics Committee (MEC-2021-0454). Patient's planning computed tomography acquired with a straight gazing angle and a 1-mm slice thickness were contoured with magnetic resonance imaging fusion when available. The organs-at-risk included the anterior segment, retina defined as a 1-mm sclera layer including the Gross Tumor Volume (GTV), eye globe, optic nerve, optic disc, and lacrimal gland. The anterior and posterior chambers, lens, ciliary body, limbus, and cornea were combined as the anterior segment. The macula was reconstructed using fundus photography.

For SRT, a dose of 50 Gy in 5 fractions was prescribed, ensuring that at least 98% of the GTV with a 2-mm isotropic expansion received 95% of the prescribed dose.¹¹ For proton planning, a 60 Gy weighted dose (relative biological effectiveness = 1.1) in 4 fractions was prescribed, ensuring that at least 95% of GTV received 90% of the prescribed dose.^{10,12,13}

Relevant dose-volume parameter values were calculated for SRT and proton plans, and risks of complications were constructed based on published dose-volume criteria.¹⁴⁻¹⁶ Profile I for maculopathy, optic-neuropathy, and visual acuity deterioration risks, included the $D_{2\%}$, D_{mean} , and $V_{30\text{Gy}}$ to the optic nerve; D_{mean} to the optic disc; and $D_{2\%}$ to the macula. Profile II of neovascular glaucoma risk included the $D_{2\%}$ and D_{mean} received by the anterior segment and the $D_{2\%}$ to the optic nerve. Profile III of radiation-induced retinopathy risk included the $D_{20\%}$ dose as well as $V_{5\text{Gy}}$, $V_{10\text{Gy}}$, $V_{20\text{Gy}}$, and $V_{30\text{Gy}}$ volumes to the retina. Profile IV of dry-eye risk included the $D_{2\%}$ and D_{mean} to the lacrimal gland.

Additional information regarding prescriptions, toxicity profiles, and treatment planning can be found in Fleury et al.^{10,17,18}

To correlate the dose-volume parameter values with individual anatomical features of patients, a machine learning model was developed using the Scikit-learn open-source library in Python v3.7 (Fig. 1). Anatomical inputs included the tumor thickness, volume, largest basal diameter, and distances to the organs-at-risk. A random forest multi-output regressor with an internal cross-validation process was selected as an estimator to lessen the statistical noise and to control the risk of overfitting when using a restricted number of patients.

The model performance was evaluated using a cross-validation with an integrated 5-Fold process. Hyperparameters *ie.*, number of decision trees, depth of each tree, and splitting properties, were optimized to provide the best-performing results for each machine learning model. Training and testing datasets were split by bootstrapping with replacement, ensuring the large anatomical variation across patients was appropriately accounted for. The following steps were repeated: (1) for generalization, 1000 random training/testing sampling scenarios were looped among (66/14) possibilities, leading to 14,000 values evaluated; and (2) for each toxicity-based Profile I to IV.

The prediction accuracy of selecting the correct treatment modality between AI-predicted and dosimetric calculated outcomes was calculated for each toxicity profile.

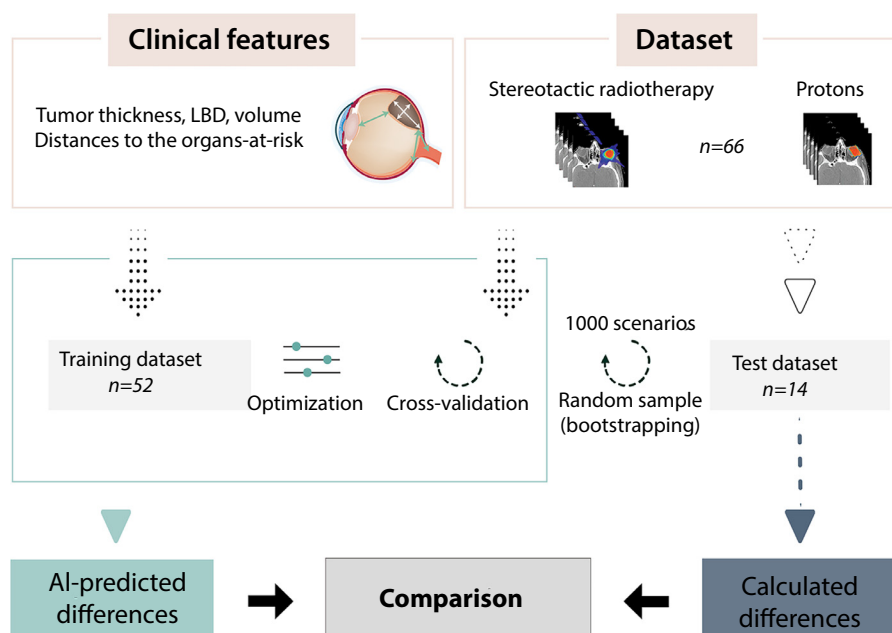


Fig. 1. Workflow for training, cross-validation process, and final comparison in the development of the AI model for dose-volume parameter value predictions. AI = artificial intelligence; LBD = largest basal diameter.

Two-sided Wilcoxon signed-rank tests were used to evaluate the statistical significance of differences between the 2 related samples using Python v3.7 and the SciPy package, and because multiple comparisons were performed, the level of significance was set to $P < 0.001$.

Results

Data of a total of 66 patients with UM were retrieved for this study, yielding 1848 dosimetric outcome points.

The boxplots in Fig. 2 show that for each dose-volume parameter, only small differences were found between AI-predicted and calculated values. Table 1 details the proportion of correct rankings of dose-volume parameter values using AI. The prediction accuracies for some important dose-volume metrics were high, like the classical $D_{2\%}$ to macula and optic nerve, at 71% and 90%, respectively. Prediction for dose-volume parameter values linked to risk of neovascular glaucoma fared less well, such as the D_{mean} to the anterior segment, which achieved only 64% accuracy. The obtained mean absolute errors for each dose-volume parameter used across all toxicity Profiles I to IV are provided in the supplementary material Table E1. The overall prediction accuracy for selecting the correct treatment modality across the entire cohort was 81%, 77%, 91%, and 93% for Profile I to IV, respectively.

Discussion

This study suggests that AI has the potential to assist in selecting and recommending the most suitable radiation

therapy technique, SRT or proton therapy, during patient evaluation in an ophthalmic oncology clinic after having required an ocular ultrasound. Based on median values, we have demonstrated in a previous study that the dosimetric comparison favored protons over SRT for the whole cohort.¹⁰ However, looking into dose-volume parameters for each patient, our previous study also showed that protons are always superior in only 14% of the cases.¹⁰ This means that, depending on which complication risk is more critical for the patient, SRT could be chosen over protons, a decision that could be discussed early in the therapeutic journey.

The decision aid described in this study could facilitate this process. Our results demonstrate that it has a fair prediction accuracy, reliably estimating dose-volume parameter values and associated complication risks between 77% and 93%. The workflow presented here follows the principle of the model-based approach used in The Netherlands for the development of proton therapy,¹⁹ which compares individual complication risks to decide on the optimal treatment technique, photons or protons. Calculating long-term complication risk involves generating treatment plans upfront for each modality, with each dosimetry voxel multiplied by the Normal Tissue Complication Probability to derive the overall complication risk. If the risk exceeds a pre-defined threshold and is deemed unacceptable with photons, then protons are justified. This approach is a long and tedious process, and the patient must wait until the simulation, planning, and comparison steps are completed before being offered treatment alternatives. In the approach proposed in this study, AI can be used almost at the time of the first consultation and after performing an ultrasound to obtain information about the location, size, and shape of the tumor

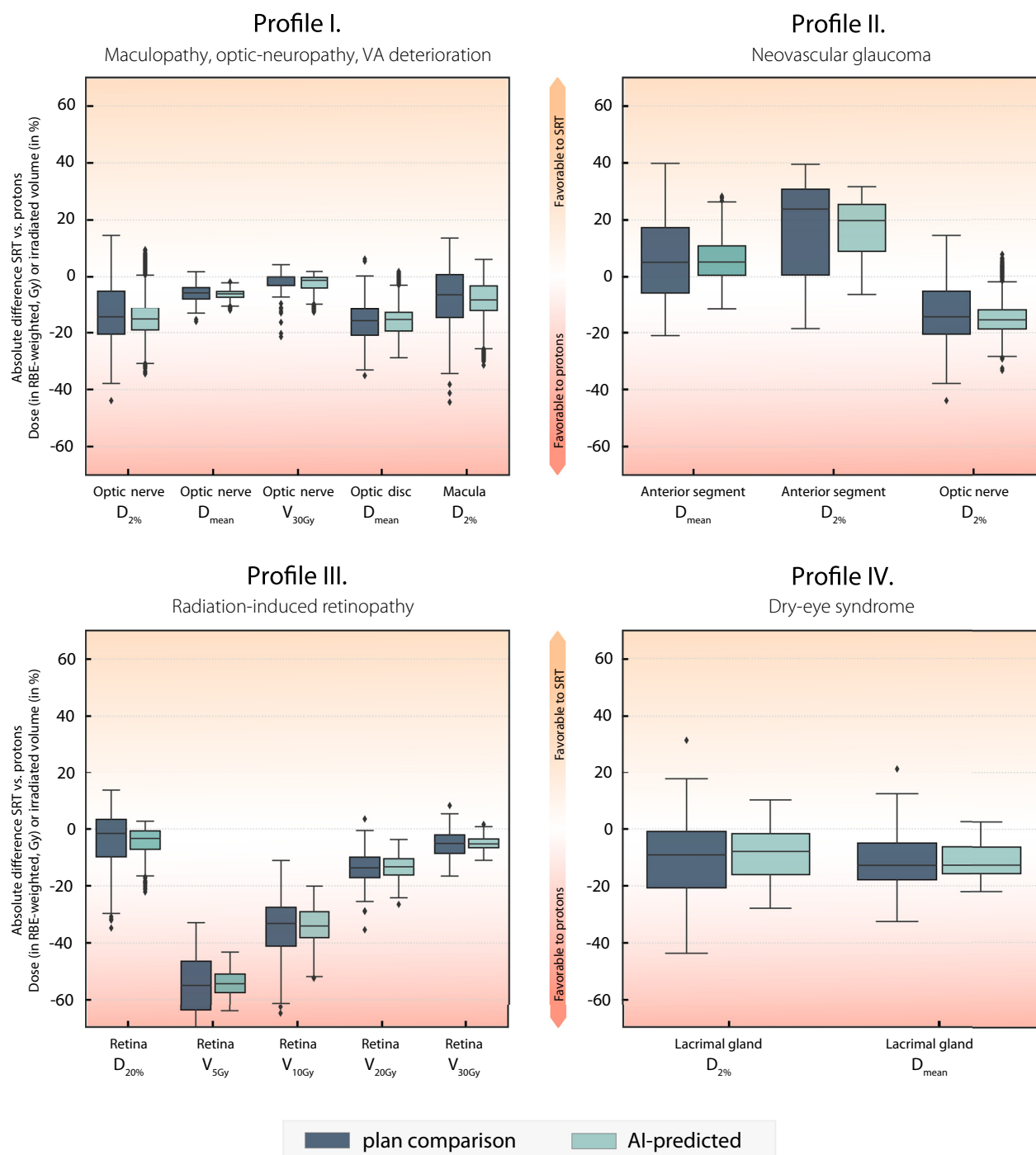


Fig. 2. AI-predicted versus calculated dose-volume parameter values for various post-treatment complication profiles. Profile I for maculopathy, optic-neuropathy, and visual acuity deterioration; Profile II for neovascular glaucoma; Profile III for retinopathy, and Profile IV for dry-eye syndrome. Negative values are favorable to proton therapy. The differences are expressed for a total treatment dose (weighted in relative biological effectiveness, Gy), or irradiated organ-at-risk volume (in % point). AI = artificial intelligence; RBE = relative biological effectiveness; VA = visual acuity.

and distances to the organs-at-risk. Using these values, the difference in planning dose-volume metrics can be estimated, allowing for the forecasting of ocular complication risks and, consequently, aiding in the selection of the optimal radiation technique.

Four types of ocular toxicity were selected to build our model, assuming they are the most clinically relevant radiation-induced complications to prevent. It is important to stress that the decision logic behind modality selection typically includes a decision 'tree', and often, a single favorable

Table 1 Toxicity-specific, machine learning-based results using 20% of the data as the test set. The process was repeated over 1000 scenarios (14,000 values evaluated) across Profiles I to IV

		Profile I	Profile II	Profile III	Profile IV
		Maculopathy, optic-neuropathy, VA deterioration	Neovascular glaucoma	Retinopathy	Dry-eye syndrome
Prediction accuracy performance	Profile-specific Dose-volume metric	81%	77%	91%	93%
Macula	D _{2%}	71%	89%		
	D _{2%}	90%			
	D _{mean}	99%			
	V _{30Gy}	53%			
Optic nerve	D _{mean} (in %)	92%	78%		
Optic disc	D _{2%}		64%		
	D _{mean}				
Anterior segment	D _{2%}			65%	
	D _{mean}				
Retina	D _{20%}			100%	
	V _{5Gy} (in %)				
	V _{10Gy} (in %)				
	V _{20Gy} (in %)				
Lacrimal gland	V _{30Gy} (in %)			99%	
	D _{2%}				
	D _{mean}				

Abbreviation: VA = visual acuity.

factor is enough to favor one modality. Additionally, the dosimetric differences between treatments are sometimes very small, suggesting that both options are somewhat equivalent. In such cases, other criteria, such as patient preferences and unsuitability for the clip placement procedure, can also play a role in treatment modality selection.

This AI approach may be further strengthened by including additional patient medical conditions, such as hypertension, diabetes,²⁰ genetics, or smoking habits. Also, while we restricted this work to two radiation therapy modalities, SRT and protons, other treatment modalities such as plaque brachytherapy or radio-frequency ablation might be added. Not all risks or doses to other organs were included in our AI tool. For example, the overall dose bath or skin dose could also generate, though much less frequently, significant side effects, including secondary cancers or skin redness. Finally, while multiple dose and fractionation regimens have been used for SRT or protons, we limited our study to the regimen used in The Netherlands.

References

1. Singh AD, Turell ME, Topham AK. Uveal melanoma: Trends in incidence, treatment, and survival. *Ophthalmology* 2011;118:1881-1885.

2. Hrbacek J, Kacperek A, Beenaker JWM, et al. PTCOG ocular statement: Expert summary of current practices and future developments in ocular proton therapy. *Int J Radiat Oncol* 2024;000(00).

3. Papakostas TD, Lane AM, Morrison M, Gragoudas ES, Kim IK. Long-term outcomes after proton beam irradiation in patients with

large choroidal melanomas. *JAMA Ophthalmol* 2017;135:1191-1196.

4. Lane AM, Kim IK, Gragoudas ES. Long-term risk of melanoma-related mortality for patients with uveal melanoma treated with proton beam therapy. *JAMA Ophthalmol* 2015;133:792-796.

5. Yazici G, Kiratli H, Ozyigit G, et al. Stereotactic radiosurgery and fractionated stereotactic radiation therapy for the treatment of uveal melanoma. *Int J Radiat Oncol Biol Phys* 2017;98:152-158.

6. Liegl R, Schmelter V, Fuerweger C, et al. Robotic cyberknife radiosurgery for the treatment of choroidal and ciliary body melanoma. *Am J Ophthalmol* 2023 2023. <https://doi.org/10.1016/j.ajo.2022.12.021>.

7. Sagoo MS, Shields CL, Mashayekhi A, et al. Plaque radiotherapy for juxtapapillary choroidal melanoma overhanging the optic disc in 141 consecutive patients. *Arch Ophthalmol* 2008;126:1515-1522.

8. Thariat J, Martel A, Matet A, et al. Non-cancer effects following ionizing irradiation involving the eye and orbit. *Cancers* 2022;14:1194.

9. Varian Medical Systems. *Planning reference guide for eclipse ocular proton planning eclipse ocular proton planning*. 2007:1-286.

10. Fleury E, Pignol JP, Kiliç E, et al. Comparison of stereotactic radiotherapy and protons for uveal melanoma patients. *Phys imaging Radiat Oncol* 2024;31. <https://doi.org/10.1016/j.phro.2024.100605>.

11. Muller K, Naus N, Nowak PJCM, et al. Fractionated stereotactic radiotherapy for uveal melanoma, late clinical results. *Radiother Oncol* 2012;102:219-224.

12. Fleury E, Trnková P, Spruijt K, et al. Characterization of the HollandPTC proton therapy beamline dedicated to uveal melanoma treatment and an interinstitutional comparison. *Med Phys* 2021;48(8):4506-4522. <https://doi.org/10.1002/mp.15024>.

13. ICRU Report 78: Prescribing, Recording, and Reporting Proton-Beam Therapy. *Int Comm Radiat Units Meas* 2007 Published online 2007.

14. Espensen CA, Kilgaard JF, Appelt AL, et al. Dose-Response and normal tissue complication probabilities after proton therapy for choroidal melanoma. *Ophthalmology* 2021;128:152-161.

15. Mishra KK, Daftari IK, Weinberg V, et al. Risk factors for neovascular glaucoma after proton beam therapy of uveal melanoma: A detailed analysis of tumor and dose-volume parameters. *Int J Radiat Oncol Biol Phys* 2013;87:330-336.

16. Daftari IK, Char DH, Verhey LJ, et al. Anterior segment sparing to reduce charged particle radiotherapy complications in uveal melanoma. *Int J Radiat Oncol Biol Phys* 1997;39:997-1010.
17. Fleury E, Trnková P, van Rij C, et al. Improving organs-at-risk sparing for choroidal melanoma patients: A CT-based two-beam strategy in ocular proton therapy with a dedicated eyeline. *Radiother Oncol* 2022;171:173-181. <https://doi.org/10.1016/j.radonc.2022.04.02>.
18. Fleury E, Trnková P, Erdal E, et al. Three-dimensional MRI-based treatment planning approach for non-invasive ocular proton therapy. *Med Phys* 2020;48(3):1315-1326. <https://doi.org/10.1002/mp.14665>.
19. Langendijk JA, Lambin P, De Ruyscher D, Widder J, Bos M, Verheij M. Selection of patients for radiotherapy with protons aiming at reduction of side effects: the model-based approach. *Radiother Oncol* 2013;107(3):267-273. <https://doi.org/10.1016/j.radonc.2013.05.007>.
20. Riechardt AI, Pilger D, Cordini D, et al. Neovascular glaucoma after proton beam therapy of choroidal melanoma: incidence and risk factors. *Graefe's Arch Clin Exp Ophthalmol* 2017;255:2263-2269.